

10. Prime Numbers

A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p .

A positive integer that is greater than 1 and is not prime is called **composite**.

Remark: The integer n is composite if and only if there exists an integer a such that a/n and $1 < a < n$.

EXAMPLE 1: The integer 7 is prime because its only positive factors are 1 and 7, whereas the integer 9 is composite because it is divisible by 3.

11. THE FUNDAMENTAL THEOREM OF ARITHMETIC:

Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of no decreasing size.

Example 2: The prime factorizations of 100, 641 , 999, and 1024 are given by

$$100 = 2.2.5.5 = 2^2.5^2$$

$$641 = 641$$

$$999 = 3.3.3.37 = 3^3.37$$

$$1024 = 2.2.2.2.2.2.2.2.2.2 = 2^{10}$$

Theorem: If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof: See book discrete mathematics, topic the fundamental theorem of arithmetic.

12. Greatest Common Divisors (GCD)

Definition: The largest integer that divides both of two integers is called the greatest common divisor of these integers.

Definition: Let a and b be integers, not both zero. The largest integer d such that $d|a$ and $d|b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $GCD(a, b)$.

EXAMPLE 1: What is the greatest common divisor of 24 and 36?

Solution: The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $GCD(24, 36) = 12$.

EXAMPLE 2: What is the greatest common divisor of 17 and 22?

Solution: The integers 17 and 22 have no positive common divisors other than 1, so that $GCD(17, 22) = 1$.

Relatively Prime:

The integers a and b are relatively prime if their greatest common divisor is 1.

EXAMPLE: By Example 2, it follows that the integers 17 and 22 are relatively prime, because $\text{GCD}(17, 22) = 1$.

Pairwise Relatively Prime

The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\text{GCD}(a_i, a_j) = 1$, whenever $1 < i < j$.

EXAMPLE: Determine whether the integers 10, 17, and 21 are pairwise relatively prime and whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\text{GCD}(10, 17) = 1$, $\text{GCD}(10, 21) = 1$, and $\text{GCD}(17, 21) = 1$, we conclude that 10, 17, and 21 are pairwise relatively prime.

13. Least Common Multiple

DEFINITION: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . The least common multiple of a and b is denoted by $lcm(a, b)$.

The least common multiple a and b is

$$lcm(a, b) = p_1^{\max(a_1, b_1)} \cdot p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}.$$

EXAMPLE: What is the least common multiple of $2^3 \cdot 3^5 \cdot 7^2$ and $2^4 \cdot 3^3$?

Solution:

$$\begin{aligned} lcm(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3) &= lcm(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3 \cdot 7^0) \\ &= 2^{\max(3,4)} \cdot 3^{\max(5,3)} \cdot 7^{\max(2,0)} \\ &= 2^4 \cdot 3^5 \cdot 7^2 \end{aligned}$$

14. Division Algorithm

The division algorithm states that for any integer a and any positive integer b there exists unique integers q and r such that

$$a = b \cdot q + r, \text{ where } 0 \leq r < b$$

We call a the dividend, b the divisor, q the quotient, and r the remainder.

Example: Let $a = 662$, $b = 414$ then

$$466 = 414 \cdot 1 + 248, \\ \text{where } q = 1, r = 248, 0 \leq r < 248$$

Remark: Division algorithm is helpful for finding greatest common divisor (GCD).

Example: Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

Solution: Successive uses of the division algorithm give

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41$$

Hence, $\text{GCD}(414, 662) = 2$, because 2 is the last nonzero remainder.

Exercise 10:

23. Use the Euclidean algorithm to find

- a) $\gcd(12, 18)$. b) $\gcd(111, 201)$.
- c) $\gcd(1001, 1331)$. d) $\gcd(12345, 54321)$.
- e) $\gcd(1000, 5040)$. f) $\gcd(9888, 6060)$.

24. Use the Euclidean algorithm to find

- a) $\gcd(1, 5)$. b) $\gcd(100, 101)$.
- c) $\gcd(123, 277)$. d) $\gcd(1529, 14039)$.
- e) $\gcd(1529, 14038)$. f) $\gcd(11111, 111111)$.

15. Congruence

Definition: If a and b are integers and $n > 0$, we write $a \equiv b \pmod{n}$ to mean $n \mid (b - a)$. We read this as “ a is congruent to b modulo (or mod) n .”

Note: $n \mid (b - a) = \frac{b-a}{n}$

Examples:

$29 \equiv 1 \pmod{7}$, and $60 \equiv 0 \pmod{15}$ are examples of congruence. The notation is used because the properties of congruence “ $\equiv$ ” are very similar to the properties of equality “ $=$ ”.

Remark: Congruence are an important and useful tool for the study of divisibility. As we shall see, they are also critical in the art of cryptography.

Theorem 15.1: For any integers a and b , and positive integer n , we have:

1. $a \equiv a \pmod{n}$. (Reflexivity)
2. If $a \equiv b \pmod{n}$ then $b \equiv a \pmod{n}$. (Symmetry)
3. If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$ (Transitivity)

OR

Show that congruence relation is an equivalence relation.

Proof:

1. Reflexivity

$$a \equiv a \pmod{n},$$

By definition

$$\begin{aligned} n &| (a - a) \\ &= n | 0 \end{aligned}$$

Since 0 is divisible by any integer.

Therefore $a \equiv a \pmod{n}$.

2. Symmetry

If $a \equiv b \pmod{n}$ then $n|(b - a)$.

Therefore, $n|(-1)(b - a)$ also $n|(a - b)$

Therefore, $b \equiv a \pmod{n}$.

3. Transitivity

If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $n|(b - a)$ and $n|(c - b)$.

Using the linear combination theorem, we have

$$\begin{aligned} n|(b - a) + (c - b) \\ = n|b - a + c - b \\ = n|(c - a) \end{aligned}$$

Thus, $a \equiv c \pmod{n}$.

By 1, 2, and 3, Hence congruence relation is an equivalence relation.

Example: Find $17^{34} \pmod{5}$

Solution:

$$\begin{aligned}17 &\equiv 2 \pmod{5} \\17^2 &\equiv 2^2 \pmod{5} \\17^2 &\equiv 4 \pmod{5} \\(17^2)^2 &\equiv 4^2 \pmod{5} \\17^4 &\equiv 1 \pmod{5} \\(17^4)^2 &\equiv 1^2 \pmod{5} \\17^8 &\equiv 1 \pmod{5} \\(17^8)^2 &\equiv 1^2 \pmod{5} \\17^{16} &\equiv 1 \pmod{5} \\(17^{16})^2 &\equiv 1^2 \pmod{5} \\17^{32} &\equiv 1 \pmod{5} \\17^{32} \cdot 17^2 &\equiv 1 \cdot 4 \pmod{5} \\17^{34} &\equiv 4 \pmod{5}\end{aligned}$$

Hence $17^{34} \pmod{5} = 4$

Exercise 12:

6. Find $2^{501} \bmod 17$.

7. Find $3^{701} \bmod 80$.